Flyback Converter Power Stage Calculation

Given Parameters

- Input Voltage: $V_{in} = 30\,\text{V}$
- Output Voltage: $V_{out} = 24\,\text{V}$
- Output Power: $P_{out} = 110\,\text{W}$
- Switching Frequency: $f_{sw} = 100\,\text{kHz}$
- Ripple Current Rate: 30% of I_{out}
- Ripple Voltage Rate: 1% of V_{out}

Calculations

1. Duty Cycle (D)

$$D = \frac{V_{out}}{V_{in}} = \frac{24}{30} = \boxed{0.8\,(80\%)}$$

2. Output Current (I_{out})

$$I_{out} = \frac{P_{out}}{V_{out}} = \frac{110}{24} \approx \boxed{4.583\,\text{A}}$$

3. Output Resistance (R)

$$R = \frac{V_{out}}{I_{out}} = \frac{24}{4.583} \approx \boxed{5.236\,\Omega}$$

4. Ripple Current (ΔI_L)

$$\Delta I_L = 30\% \times I_{out} = 0.3 \times 4.583 \approx \boxed{1.375\,\text{A (peak-to-peak)}}$$

5. Maximum/Minimum Inductor Current

$$I_{L_{\max}} = I_{out} + \frac{\Delta I_L}{2} = 4.583 + 0.6875 \approx \boxed{5.27\,\text{A}}$$

$$I_{L_{\min}} = I_{out} - \frac{\Delta I_L}{2} = 4.583 - 0.6875 \approx \boxed{3.896\,\text{A}}$$

6. Ripple Voltage (ΔV_{out})

$$\Delta V_{out} = 1\% \times V_{out} = 0.01 \times 24 = \boxed{0.24\,\text{V (peak-to-peak)}}$$

7. **Output Capacitor** (C)

$$C = \frac{I_{out} \cdot D}{f_{sw} \cdot \Delta V_{out}} = \frac{4.583 \cdot 0.8}{100,000 \cdot 0.24} \approx \boxed{152.8\,\mu\text{F}}$$

8. **Inductor** (L)

$$L = \frac{(V_{in} - V_{out}) \cdot D}{f_{sw} \cdot \Delta I_L} = \frac{(30 - 24) \cdot 0.8}{100,000 \cdot 1.375} \approx \boxed{34.9\,\mu\text{H}}$$

9. **MOSFET Selection**

- Voltage Rating: $\geq 40\,\text{V}$
- Current Rating: $\geq 8\,\text{A}$

10. **Diode Selection**

- Voltage Rating: $\geq 40\,\text{V}$
- Current Rating: $\geq 5\,\text{A}$

Summary Table

Parameter	**Value**
Duty Cycle	80% (0.8)
Output Current	4.583 A
Output Resistance	5.236 Ω
Ripple Current	1.375 A (p-p)
Inductor Current (Max/Min)	5.27 A/3.896 A
Ripple Voltage	0.24 V (p-p)
Output Capacitor	152.8 μF
Inductor	34.9 μH

Notes

- The calculations assume a **buck converter topology**, though the title mentions a **flyback converter**. Flyback designs require additional parameters (e.g., transformer turns ratio).
- Thermal management is critical for MOSFET and diode.